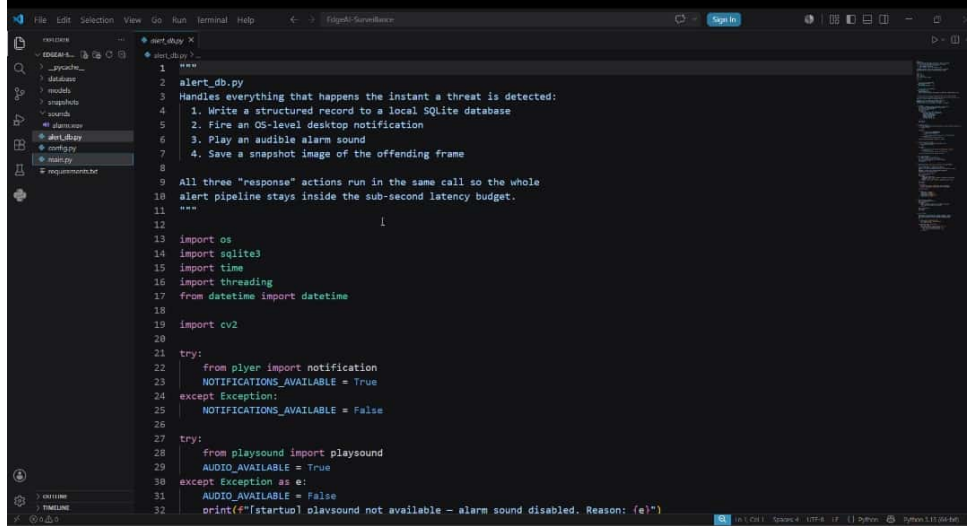


15:42

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The screenshot shows a mobile phone screen with a code editor application open. The status bar at the top displays the time 15:42 and various system icons including network, battery, and signal strength. The code editor has a dark theme and a sidebar on the left showing a file tree. The main editor area contains Python code for an alert system. The code includes comments, imports, and exception handling for notifications and audio playback.

```
1 """
2 alert_db.py
3 Handles everything that happens the instant a threat is detected:
4 1. Write a structured record to a local SQLite database
5 2. Fire an OS-level desktop notification
6 3. Play an audible alarm sound
7 4. Save a snapshot image of the offending frame
8
9 All three "response" actions run in the same call so the whole
10 alert pipeline stays inside the sub-second latency budget.
11 """
12
13 import os
14 import sqlite3
15 import time
16 import threading
17 from datetime import datetime
18
19 import cv2
20
21 try:
22     from plyer import notification
23     NOTIFICATIONS_AVAILABLE = True
24 except Exception:
25     NOTIFICATIONS_AVAILABLE = False
26
27 try:
28     from playsound import playsound
29     AUDIO_AVAILABLE = True
30 except Exception as e:
31     AUDIO_AVAILABLE = False
32     print(f"(!startuo) playsound not available - alarm sound disabled. Reason: {e}")
```

15:42

VoLTE 93%

```
File Edit Selection View Go Run Terminal Help -> File Explorer  
alert_db.py X  
35 # Use the script's own folder as the base, NOT the current working directory.  
36 # VS Code (Run button / debugger) sometimes launches with a different cwd than  
37 # a plain terminal does, which silently breaks relative paths like "sounds/alarm.wav".  
38 BASE_DIR = os.path.dirname(os.path.abspath(__file__))  
39  
40 DB_PATH = os.path.join(BASE_DIR, "database", "logs.db")  
41 SNAPSHOT_DIR = os.path.join(BASE_DIR, "snapshots")  
42 ALARM_SOUND_PATH = os.path.join(BASE_DIR, "sounds", "alarm.wav")  
43  
44  
45 def init_db():  
46     """Create the database + table if they don't already exist."""  
47     os.makedirs(os.path.dirname(DB_PATH), exist_ok=True)  
48     conn = sqlite3.connect(DB_PATH)  
49     cur = conn.cursor()  
50     cur.execute("""  
51         CREATE TABLE IF NOT EXISTS detection_log (  
52             id INTEGER PRIMARY KEY AUTOINCREMENT,  
53             timestamp TEXT NOT NULL,  
54             object_class TEXT NOT NULL,  
55             confidence REAL NOT NULL,  
56             snapshot_path TEXT,  
57             alert_type TEXT NOT NULL  
58         )  
59     """)  
60     conn.commit()  
61     conn.close()  
62  
63  
64 def play_alarm_async():  
65     """Play the alarm sound on a background thread so it never blocks the video loop."""
```

```
103 def log_detection(timestamp, object_class, confidence, snapshot_path, alert_type),
117     VALUES (t, f, f, f, f)"""
118     (timestamp, object_class, confidence, snapshot_path, alert_type),
119 )
120 conn.commit()
121 conn.close()
122
123 _notify_async(
124     title=f"⚠ {alert_type}: {object_class.upper()} detected",
125     message=f"Confidence {confidence*100:.1f}% at {timestamp}",
126 )
127 _play_alarm_async()
128
129 return {
130     "timestamp": timestamp,
131     "object_class": object_class,
132     "confidence": confidence,
133     "snapshot_path": snapshot_path,
134     "alert_type": alert_type,
135 }
136
137
138 def get_recent_logs(limit=10):
139     conn = sqlite3.connect(DB_PATH)
140     cur = conn.cursor()
141     cur.execute(
142         "SELECT timestamp, object_class, confidence, alert_type "
143         "FROM detection_log ORDER BY id DESC LIMIT ?",
144         (limit,),
145     )
146     rows = cur.fetchall()
147     conn.close()
```

```
File Edit Selection View Go Run Terminal Help -> | Edge AI Surveillance | Sign In |
alertLibray X
149
150
151
152 class AlertCooldown:
153     """
154     Prevents alert spam. Without this, a person standing in frame for
155     10 seconds at 20 FPS would create ~200 log rows and 200 popups.
156     Only allows one alert per object class every 'cooldown_seconds'.
157     """
158     def __init__(self, cooldown_seconds=5):
159         self.cooldown_seconds = cooldown_seconds
160         self._last_alert_time = {}
161
162     def should_alert(self, object_class):
163         now = time.time()
164         last = self._last_alert_time.get(object_class, 0)
165         if now - last >= self.cooldown_seconds:
166             self._last_alert_time[object_class] = now
167             return True
168         return False
169
```

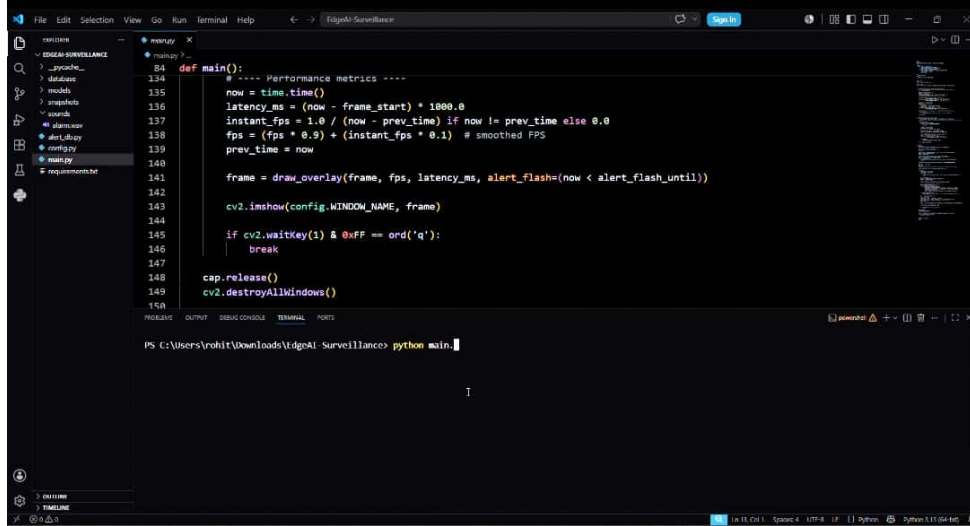
```
File Edit Selection View Go Run Terminal Help Edge-AI Surveillance
config.py
1 """
2 config.py
3 All the "knobs" for the project live here so you don't have to
4 dig through main.py to tweak behaviour for your demo.
5 """
6
7 # ---- Video source ----
8 # 0 = default webcam
9 # "video.mp4" = a local video file (good for a reliable demo if no camera available)
10 # "rtsp://user:pass@ip:port/stream" = an IP camera / CCTV RTSP feed
11 VIDEO_SOURCE = 0
12
13 # ---- Model ----
14 MODEL_PATH = "models/yolov8n.pt" # auto-downloads on first run if not present
15 CONFIDENCE_THRESHOLD = 0.45
16
17 # ---- Threat classes ----
18 # COCO class names that should trigger a HIGH priority alert.
19 # (COCO has no "drone" class - add a custom-trained model later if you need that.)
20 THREAT_CLASSES = {"person", "car", "truck", "motorcycle", "bicycle"}
21
22 # ---- Alerting ----
23 ALERT_COOLDOWN_SECONDS = 5 # min seconds between repeat alerts for the same class
24
25 # ---- Display ----
26 WINDOW_NAME = "Edge-AI Surveillance Dashboard"
27 SHOW_FPS_OVERLAY = True
28
```

```
File Edit Selection View Go Run Terminal Help - Edge-Surveillance
Sign In

main.py
28 def draw_overlay(frame, fps, latency_ms, alert_flash):
29     overlay = frame.copy()
30     cv2.rectangle(overlay, (0, 0), (w, 40), (20, 20, 20), -1)
31     cv2.addWeighted(overlay, 0.6, frame, 0.4, 0, frame)
32
33     cv2.putText(frame, f"FPS: {fps:.1f}", (10, 27),
34                 cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0), 2)
35     cv2.putText(frame, f"Latency: {latency_ms:.1f} ms", (150, 27),
36                 cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0), 2)
37
38     status_text = "STATUS: LIVE"
39     status_color = (0, 255, 0)
40     if alert_flash:
41         status_text = "STATUS: THREAT DETECTED"
42         status_color = (0, 0, 255)
43     cv2.putText(frame, status_text, (w - 320, 27),
44                 cv2.FONT_HERSHEY_SIMPLEX, 0.65, status_color, 2)
45
46     # Red border flash on active alert - visible even from across a room
47     if alert_flash:
48         cv2.rectangle(frame, (0, 0), (w - 1, h - 1), (0, 0, 255), 6)
49
50     return frame
51
52
53 def draw_detections(frame, boxes, names):
54     """Draw bounding boxes + class label + confidence for every detection."""
55     threat_hits = [] # list of (class_name, confidence) that count as threats this frame
56
57     for box in boxes:
58         conf = float(box.conf[0])
```

```
File Edit Selection View Go Run Terminal Help  Edge AI-Surveillance
main.py
84 def main():
85     print("[init] opening video source: (config.VIDEO_SOURCE)")
86     cap = cv2.VideoCapture(config.VIDEO_SOURCE)
87     if not cap.isOpened():
88         print("[error] Could not open video source. Check config.VIDEO_SOURCE.")
89         return
90
91     cooldown = AlertCooldown(cooldown_seconds=config.ALERT_COOLDOWN_SECONDS)
92
93     prev_time = time.time()
94     fps = 0.0
95     alert_flash_until = 0 # timestamp until which the red flash stays on screen
96     print("[run] Press 'q' in the video window to quit.\n")
97
98     while True:
99         frame_start = time.time()
100         ret, frame = cap.read()
101         if not ret:
102             print("[warn] frame grab failed - end of stream or camera disconnected.")
103             break
104
105         # ---- Inference ----
106         results = model.predict(frame, verbose=False)
107         boxes = results[0].boxes
108
109         frame, threat_hits = draw_detections(frame, boxes, names)
110
111         # ---- Threat decision + alert pipeline ----
112         alert_fired_this_frame = False
113         for cls_name, conf in threat_hits:
114             if cooldown.should_alert(cls_name):
```

```
File Edit Selection View Go Run Terminal Help Edge-Surveillance
main.py
84 def main():
85     # --- Performance metrics ---
86     now = time.time()
87     latency_ms = (now - frame_start) * 1000.0
88     instant_fps = 1.0 / (now - prev_time) if now != prev_time else 0.0
89     fps = (fps * 0.9) + (instant_fps * 0.1) # smoothed FPS
90     prev_time = now
91
92     frame = draw_overlay(frame, fps, latency_ms, alert_flash=(now < alert_flash_until))
93
94     cv2.imshow(config.WINDOW_NAME, frame)
95
96     if cv2.waitKey(1) & 0xFF == ord('q'):
97         break
98
99     cap.release()
100     cv2.destroyAllWindows()
101
102     print("\n[summary] Last 10 detections logged this session (or earlier):")
103     for row in get_recent_logs(10):
104         print(" ", row)
105
106 if __name__ == "__main__":
107     main()
108
```

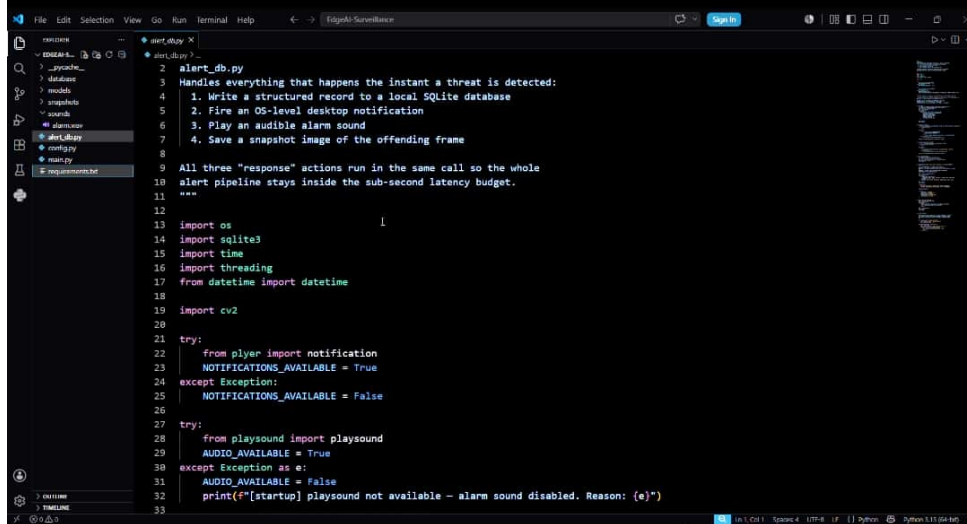


```
File Edit Selection View Go Run Terminal Help EdgeAI-Surveillance
main.py x
main.py > ...
134 def main():
135     # ---- Performance metrics ----
136     now = time.time()
137     latency_ms = (now - frame_start) * 1000.0
138     instant_fps = 1.0 / (now - prev_time) if now != prev_time else 0.0
139     fps = (fps * 0.9) + (instant_fps * 0.1) # smoothed FPS
140     prev_time = now
141
142     frame = draw_overlay(frame, fps, latency_ms, alert_flash=(now < alert_flash_until))
143
144     cv2.imshow(config.WINDOW_NAME, frame)
145
146     if cv2.waitKey(1) & 0xFF == ord('q'):
147         break
148
149     cap.release()
150     cv2.destroyAllWindows()
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15:43

N 🔊 🔋 🔴 VoLTE 📶 93%



The screenshot shows a mobile phone screen with a code editor application open. The status bar at the top displays the time 15:43, various icons (signal, battery, VoLTE), and a 93% battery level. The code editor has a dark theme and a sidebar on the left showing a file tree. The main editor area displays Python code for a module named `alert_db.py`. The code includes comments, imports, and logic for handling alerts, such as writing to a SQLite database, sending notifications, and playing an alarm sound. The code is as follows:

```
1 alert_db.py
2 Handles everything that happens the instant a threat is detected:
3 1. Write a structured record to a local SQLite database
4 2. Fire an OS-level desktop notification
5 3. Play an audible alarm sound
6 4. Save a snapshot image of the offending frame
7
8
9 All three "response" actions run in the same call so the whole
10 alert pipeline stays inside the sub-second latency budget.
11 """
12
13 import os
14 import sqlite3
15 import time
16 import threading
17 from datetime import datetime
18
19 import cv2
20
21 try:
22     from plyer import notification
23     NOTIFICATIONS_AVAILABLE = True
24 except Exception:
25     NOTIFICATIONS_AVAILABLE = False
26
27 try:
28     from playsound import playsound
29     AUDIO_AVAILABLE = True
30 except Exception as e:
31     AUDIO_AVAILABLE = False
32     print(f"[startup] playsound not available - alarm sound disabled. Reason: {e}")
33
```